

Q3 (Mid Sem)

Fitted regression line: $\hat{y} = 556.26 - 4.27x$ on 38 pairs of observations where 556.26 and -4.27 are the least square estimators. The residual standard error is 36.37. The sample mean and standard deviation of x are 169.1 and 11.17 respectively.

- For the value 170 of x , what is the estimate of the expected value of y under the simple linear regression model?
- What is the standard error of the estimator in part (a)? Derive the expression for the standard error and report the numerical value.
- For the value 170 of x , what is the predicted value of y ?
- What is the standard error of this prediction?

Solⁿ →

model: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$; $\epsilon_i \sim N(0, \sigma^2)$, $i = 1, 2, \dots, n$

OLS estimators:
$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum (x_i - \bar{x})y_i}{S_{xx}} = \sum c_i y_i, \quad S_{xx} = \sum (x_i - \bar{x})^2$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \sum d_i y_i, \quad d_i = \frac{1}{n} - \bar{x} c_i, \quad c_i = \frac{(x_i - \bar{x})}{S_{xx}}$$

$E(\hat{\beta}_0) = \beta_0$; $E(\hat{\beta}_1) = \beta_1$

$\text{Var}(\hat{\beta}_1) = \text{Var}(\sum c_i y_i) = \sum c_i^2 \text{Var}(y_i) = \frac{\sigma^2}{S_{xx}}$

$SE(\hat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}}$

$\text{Var}(\hat{\beta}_0) = \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x})$

$$= \text{Var}(\bar{y}) + \text{Var}(\hat{\beta}_1 \bar{x}) - 2 \text{Cov}(\bar{y}, \hat{\beta}_1 \bar{x}) = \frac{\sigma^2}{n} + (\bar{x})^2 \frac{\sigma^2}{S_{xx}} - 2 \bar{x} \frac{\sigma^2}{S_{xx}}$$

$$= \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \sigma^2$$

$\therefore \text{Cov}(\bar{y}, \hat{\beta}_1) = 0$

$SE(\hat{\beta}_0) = \sigma \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$

$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \text{Cov}(\sum d_i y_i, \sum c_i y_i) = \sum_{i,j} d_i c_j \text{Cov}(y_i, y_j) = \sum_i d_i c_i \sigma^2 = -\frac{\bar{x} \sigma^2}{S_{xx}}$

Fitted mean: $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$

$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 = \frac{1}{n} \sum y_i + \bar{x} \sum c_i y_i = \sum \left(\frac{1}{n} + \bar{x} c_i \right) y_i = \sum \hat{e}_i y_i$

① $n = 38$; $\hat{\beta}_0 = 556.26$, $\hat{\beta}_1 = -4.27$, $s = 36.37$; $\bar{x} = 169.1$, $S_{xx} = (n-1)s_x^2$, $s_x = 11.17$

$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 = 556.26 + (-4.27) \times 170 = 169.64$

$S_x^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

② $\text{Var}(\hat{y}_0) = \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \text{Var}(\hat{\beta}_0) + x_0^2 \text{Var}(\hat{\beta}_1) + 2x_0 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$

$$= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) + x_0^2 \frac{\sigma^2}{S_{xx}} + 2x_0 \left(-\frac{\bar{x} \sigma^2}{S_{xx}} \right) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

$SE(\hat{y}_0) = \sigma \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \approx s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$

$S_{xx} = (n-1)s_x^2 = 37 \times (11.17)^2 = 4616.64$

$\therefore SE(\hat{y}_0) \approx 5.92$

Include both
 (i) uncertainty in estimating the regression line
 (ii) the natural random variation of individual observation around the line.

③ same as ②

④ $y_{\text{new}} = \beta_0 + \beta_1 x_0 + \epsilon_{\text{new}}$ (ϵ_{new} independent of the sample estimator)
 point prediction: $\hat{y}_{\text{new}} = \hat{\beta}_0 + \hat{\beta}_1 x_0$

Variance of a new individual prediction at x_0 means that how much the actual new y may vary around our fitted line at x_0 .

$\text{Var}(y_{\text{new}} | \text{prediction at } x_0) = \text{Var}(y_{\text{new}} - \hat{y}_0) = \text{Var}(\epsilon_{\text{new}}) + \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0)$

$$= \sigma^2 + \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

$SE(y_{\text{new}} - \hat{y}_0) = \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \approx s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} = 36.26 \sqrt{1.0269} \approx 36.84$

Q1:- The model is given by

$$y_i = \beta x_i + \epsilon_i x_i$$

we can write $\epsilon_i = \frac{y_i - \beta x_i}{x_i}$

The sum of squared errors is given by

$$SSE = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n \left(\frac{y_i - \beta x_i}{x_i} \right)^2$$

Now, derivative of SSE with respect to β is taken and set to zero.

$$\frac{d SSE}{d \beta} = 0$$

$$\Rightarrow \sum_{i=1}^n \left(\frac{y_i}{x_i} - \beta \right) = 0$$

$$\Rightarrow \sum_{i=1}^n \frac{y_i}{x_i} - n\beta = 0$$

$$\Rightarrow \beta = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{x_i}$$

(a) Least squares estimator: $\beta_{LS} = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{x_i}$

(b) The pdf of ϵ_i is $f(u) = \frac{1}{2\lambda} \exp\left(-\frac{|u|}{\lambda}\right)$, $-\infty < u < \infty$

Now, substituting $u = \frac{y_i - \beta x_i}{x_i}$, the joint likelihood function is given by.

$$L(\beta, \lambda) = \left(\frac{1}{2\lambda} \right)^n \exp\left(- \sum_{i=1}^n \frac{|y_i - \beta x_i|}{\lambda} \right).$$

The log-likelihood function is given by.

$$\ell(\beta, \lambda) = -n \log(2\lambda) - \frac{1}{\lambda} \sum_{i=1}^n \left| \frac{y_i - \beta x_i}{x_i} \right|$$

To find the maximum likelihood estimator of β , the log-likelihood

function is maximized with respect to β .

This is equivalent to minimizing the term $\sum_{i=1}^n \left| \frac{y_i - \beta x_i}{x_i} \right|$.

Let, $t_i = \frac{y_i}{x_i}, i=1, 2, \dots, n$. then ~~we~~ we ~~maximize~~ minimize

$$\sum_{i=1}^n |t_i - \beta|.$$

~~we know that~~ Now the value of β that minimizes the sum of absolute deviations is the sample median of t_i values.

$\therefore$ Maximum likelihood estimator: $\hat{\beta}_{ML} = \text{median}(t_i)_{i=1, \dots, n}$
 $= \text{median}\left(\frac{y_1}{x_1}, \dots, \frac{y_n}{x_n}\right).$

BSDS midterm Soln

3 a) The model "final" is

$$\text{Total}_i = \beta_1 Q1_i + \beta_2 \text{Attendance}_i + \beta_3 \text{Midterm}_i + \epsilon_i$$

$i = 1, \dots, n$

$$E(\epsilon_i) = 0 \quad \text{Var}(\epsilon_i) = \sigma^2, \quad \epsilon_i \text{ indep.}$$

(b)

Predictor	Coefficient	Standard Error
Q1	0.5867	0.2057
Attendance	0.8700	0.2948
Midterm	1.3260	0.1189

(c) bic has been used.

The choice of bic is indicated by sbc in the R output.

For being selected as the optimal model, bic should be lowest among all models.

The chosen model "final" indeed has the lowest bic value as indicated by the values in the sbc column.

(d) adjr $\rightarrow$ need highest $\rightarrow$ Model ⁽⁴⁾ $Q_1 + Q_2 + \text{Attendance} + \text{Midterm}$

$C_p \rightarrow$ need closest to $p+1 \rightarrow$ The full model (5) has

$$p = 5 \text{ and } C_p = 6.0$$

$$Q_1 + Q_2 + Q_3 + \text{Attendance} + \text{Midterm}$$

aic $\rightarrow$ need lowest $\rightarrow$ Model (4) $Q_1 + Q_2 + \text{Attendance} + \text{Midterm}$

bic $\rightarrow$ need lowest $\rightarrow$ Model (3) $Q_1 + \text{Attendance} + \text{Midterm}$

(e) "-1" tells the function to drop the intercept term.

The coefficient of the intercept term is insignificant with p-value 0.9157 in the full model,

So it is increasing the required number of parameters to be estimated, without significantly improving the quality of the fit. Therefore, it is advisable to drop this term.

$$\begin{aligned}
 4 \text{ (a)} \quad \hat{Y} &= X \hat{\beta} = X(X^T X)^{-1} X^T Y \\
 &= X(X^T X)^{-1} X^T [X \beta + E] \\
 &= X(X^T X)^{-1} X^T X \beta + X(X^T X)^{-1} X^T E \\
 &= X \beta + X(X^T X)^{-1} X^T E = X \beta + H E \quad \text{where } H = X(X^T X)^{-1} X^T
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(\hat{Y}) &= H \text{Var}(E) H^T \\
 &= X(X^T X)^{-1} X^T \sigma^2 I X(X^T X)^{-1} X^T \quad [\text{Var } E = \sigma^2 I] \\
 &= \sigma^2 X(X^T X)^{-1} X^T
 \end{aligned}$$

$$\begin{aligned}
 \sum_{i=1}^n \text{Var}(\hat{y}_i) &= \text{tr}(\text{Var} \hat{Y}) \\
 &= \sigma^2 \text{tr}(X(X^T X)^{-1} X^T) \\
 &= \sigma^2 \text{tr}[X^T X (X^T X)^{-1}] \\
 &= \sigma^2 \text{tr}(I_{p+1}) \quad \text{where } I_{p+1} \text{ is identity matrix} \\
 &\quad \text{with dim } (p+1) \times (p+1) \\
 &= \sigma^2 (p+1).
 \end{aligned}$$

(b) We need $(X^T X)$ to be of full rank $(p+1)$, that is, we need the columns of X to be linearly independent.

(c) The quantity $\sum_{i=1}^n \text{Var}(\hat{y}_i)$ increases linearly with the number of predictors. So including more predictors increases the sum of variance of the fitted values, which is undesirable.